

FIELDNODE · DESIGN DECISIONS REGISTER

FieldNode design decisions register

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DOCUMENT CONTROL

Document ID	FND-DEC-001	Version	0.4
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Value engineering updated after the approved follow-ups: USD 139.50 base node, USD 148.50 with the shield (programming header on line 7)
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 3 (FND-DDR-001, O1 to O3); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened; open decisions moved out of the build plan

FieldNode design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The solar panel frame has a flat back lip at least 12 mm wide	The panel clips bolt through it; without it the clip position changes	FND-DDR-003
2	The enclosure model, its boss spacing and its lug kit	They set the internal plate holes and the lug positions	FND-DDR-003
3	The band clamp torque that gives 1,000 N of preload	The calculation note assumes that preload	FND-CAL-001

■ Value engineering

Value-engineering target: USD 150 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 139.50 (USD 10.50 under the target) for the base node; USD 148.50 with the sun shield (USD 1.50 under the target). Main cost drivers and savings worth trying:

- The largest lines are the power board (USD 22.50, a prototype carrier price including the USD 0.50 serial programming header added for the 2026-10-02 firmware update decision; the board design is TRL 4 work), the enclosure with vent and lug kit (USD 20), the solar panel (USD 14), the controller and LoRa module (USD 14) and the antenna (USD 10).
- Making the design constructable repriced lines 1, 5, 13 and 14 and added line 15, the plug-in connectors and rail fuses (USD 5); the base node rose from USD 126 to USD 139. The programming header (2026-10-02) adds USD 0.50; the 915 MHz whip costs the same as the band-specific whip it names.
- Savings worth trying: the sun shield (USD 9) is an option for hot-climate sites only, so the base node carries none; the power board price should fall once its design is done.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D12 (LoRaWAN, STM32WL-class module, one 6 Ah LiFePO4 cell, 6 W panel, two M12 ports, 15 min interval and others)	Amish: "i accept all your recommendations, go with them across all repos."	FND-DDR-001, FND-DDR-002
2026-09-25	Sun shield for hot-climate sites only; base node mass limit 2.5 kg; 9 V class panel; 100 mW sensor allowance; airtime rule at slow spreading factors	Amish: go with recommendation	FND-DDR-002
2026-09-30	Design for construction: V-blocks, bottom-face layout, bracket, shield fixing, connector strip, enclosure lugs and other changes that make the node buildable	Amish: "i accept your recommended changes on design that are currently being sent across for my approval"	FND-DDR-003
2026-09-30	Thumb screws on the shield for the prototype; 5 mm spacers for wall mounting; accept the 0.05 kg mass margin and weigh at TRL 4; fit the shield on the first prototype	Amish, same instruction	FND-DDR-003, A1 to A4
2026-10-02	Pilot region and radio band: US915 (North America) is the default first variant with a 915 MHz whip; the band switches to that of the first adopting project's site if it is outside North America (EU868, AS923 or IN865)	Amish: "i approve your recommendations for all 555 open decisions."	FND-DDR-001, O1
2026-10-02	Sensor port pin assignment: the candidate pinout of FND-CAL-001 is the proposed standard (pin 1 switched rail, pin 2 data A, pin 3 ground, pin 4 data B, pin 5 analog), sent to HeatMap Node and the next adopting project for sign-off	Amish: "i approve your recommendations for all 555 open decisions."	FND-DDR-001, O2
2026-10-02	Firmware update method in the field: by cable inside the box with the lid open (a USB or serial header on the board, no extra hole in the enclosure); over-the-air updates left for a later private-gateway variant	Amish: "i approve your recommendations for all 555 open decisions."	FND-DDR-001, O3